Application Handover, Completion & AI Agent Integration Plan

1. Project Description

The objective of this project is to take the existing application from its current frontend/UI-complete stage and transform it into a fully functional, stable, and production-ready application.

The existing frontend will be reviewed and properly connected with the backend services, APIs, database, authentication, storage, and AI functionality.

A key part of the updated scope is the implementation and integration of multiple specialized AI Agents. Instead of relying only on one general AI Assistant, the platform will communicate with the appropriate AI Agent depending on the user's selected service or request.

Examples include:

- Automobile Sales AI Agent
- Parts & Accessories AI Agent
- Roadside Assistance AI Agent
- Additional specialized AI Agents defined for other application modules

The backend will manage the communication between the application and these agents, ensuring that the correct API/agent is called when required.

2. Project Scope

The development scope will include:

- Review existing frontend and backend codebase
- Review current database and API structure
- Frontend-to-backend API integration
- Database connectivity and data persistence
- User registration and login
- Role-based onboarding for:
    - Dealer
    - Member
    - Specialist
    - Runner
- Google authentication integration/fixes
- Apple authentication integration/fixes
- Image/file upload and storage
- Form validation and error handling

- Existing AI Assistant troubleshooting
- **Specialized AI Agent integration**
- **AI Agent API routing and communication**
- **Passing relevant user/module context to the correct AI Agent**
- **Receiving and displaying AI Agent responses**
- AI API error and fallback handling
- Loading, success, and error states
- API response and JSON error resolution
- Complete end-to-end testing
- Bug fixing and stabilization
- Production deployment verification
- Source-code and technical handover

The original proposal already includes AI Assistant integration, API/backend connectivity, database functionality, authentication, and complete end-to-end testing; the updated scope expands that AI component into a multi-agent architecture.

3. Development Approach

Phase 1 – Technical Audit & Environment Setup

We will first review the complete existing application and understand how all technical components are currently connected.

The audit will cover:

- Frontend codebase
- Backend architecture
- Existing APIs/endpoints
- Database configuration
- Authentication
- Environment variables
- Image/file storage
- Google and Apple authentication
- Existing AI Assistant/API configuration
- Available AI Agents and their endpoints
- Existing deployment environment

For the AI system specifically, we will identify:

- Which AI Agents already exist
- Which application module should communicate with each agent
- Required API requests and responses
- Authentication/API keys required for each agent
- Data/context required by each agent
- Existing AI errors or incomplete integrations

This extends the technical audit already defined in the original plan, which includes checking backend architecture, API endpoints, database, authentication, storage, AI service configuration, and deployment.

Phase 2 – Backend, Database & API Integration

Once the technical audit is completed, we will stabilize the communication between:

Frontend → Backend/API → Database → Response → Frontend

We will verify and fix:

- API requests
- API responses
- Database read/write operations
- User-data persistence
- Role-based permissions
- Authentication tokens/sessions
- JSON response handling
- Backend error handling
- Missing or broken endpoints

Existing issues such as **“Unexpected end of JSON input”** will be traced to their backend/API source and resolved.

Phase 3 – Authentication & Core User Flows

After stabilizing the backend foundation, we will complete the major user flows.

This includes:

- Registration
- Login
- Dealer onboarding
- Member onboarding
- Specialist onboarding
- Runner onboarding
- Google sign-in
- Apple sign-in
- Role-based access
- Role-based navigation
- Image uploads
- Form submissions
- Database updates

Each workflow will be tested end-to-end to ensure that the frontend correctly communicates with the backend and database.

4. AI Agent Architecture & Integration

This is an important part of the updated scope.

The application will support **multiple specialized AI Agents**, with each agent responsible for its relevant domain.

For example:

Automobile Sales AI Agent
Handles AI requests related to automobile sales and the relevant sales functionality.

Parts & Accessories AI Agent
Handles requests relating to automotive parts and accessories.

Roadside Assistance AI Agent
Handles requests related to roadside-assistance functionality.

Additional AI Agents
Other specialized agents can be connected according to the modules and APIs defined within the existing application architecture.

AI Agent Request Flow

The expected communication flow will be:

User Request → Application → Backend → Agent Routing → Relevant AI Agent/API → Backend → Application → User

The backend will determine or receive the appropriate agent context and communicate with the corresponding AI service.

AI Integration Work

The development will include:

- Review of existing AI APIs/agents
- Integration of required AI Agent endpoints
- Agent-specific API configuration
- Backend routing/orchestration
- Passing relevant request context
- Processing AI responses
- Returning responses to the correct application module
- Loading states during AI processing
- AI timeout/error handling
- Invalid-response handling
- Fallback handling when an AI service is unavailable
- Fixing the existing **“Something went wrong”** AI Assistant issue

- End-to-end testing of each supported AI Agent

The existing proposal currently covers fixing the AI Assistant and resolving its errors; this updated section makes the specialized agent requirement explicit.

5. Validation & Error Handling

Proper validation and user feedback will be implemented throughout the application.

This includes:

- Required fields
- Email validation
- Password validation
- Invalid input formats
- Clear validation messages
- API error messages
- AI Agent error messages
- Loading indicators
- Success confirmations
- Failed request handling
- Network error handling
- Invalid AI/API response handling

The goal is to ensure users always understand whether an operation is processing, completed successfully, or has encountered an issue.

6. Milestone-Based Development Plan

Milestone 1 – Technical Audit & Backend Foundation- $300

Scope

- Review existing frontend/backend
- Review database structure
- Review APIs/endpoints
- Verify environment configuration
- Review authentication architecture
- Review existing deployment
- Review current AI Assistant setup
- Review available AI Agents/APIs
- Identify agent-to-module relationships
- Identify missing/broken backend integrations

- Stabilize frontend-to-backend communication
- Resolve core API/JSON issues

Deliverable

A stable technical foundation with clear backend, database, API, and AI Agent architecture ready for implementation.

Milestone 2 – Authentication & Role-Based User Flows- $600

Scope

- User registration
- User login
- Dealer registration/login
- Member registration/login
- Specialist registration/login
- Runner registration/login
- Google authentication
- Apple authentication
- Session/token management
- Role-based access
- Role-based redirects/navigation
- Registration data storage/retrieval

Deliverable

Fully functional authentication and onboarding workflows for all supported user roles.

The structure remains aligned with the original Milestone 2 scope.

Milestone 3 – Core Features, Image Uploads & AI Agent Integration- $600

Scope

- Fix image/file uploads
- Connect storage with backend/database
- Fix existing AI Assistant
- Resolve existing AI errors
- Integrate specialized AI Agents
- Automobile Sales AI Agent integration
- Parts & Accessories AI Agent integration
- Roadside Assistance AI Agent integration

- Integrate other defined agents where applicable
- Implement backend AI Agent routing
- Connect relevant modules with their corresponding agents
- Pass appropriate request/context data
- Process AI responses
- AI loading/error/fallback handling
- Complete remaining backend-dependent features
- Form submissions and database updates
- Proper form validation
- Loading/success/error states
- API failure handling

Deliverable

A functional application where the major frontend and backend features operate end-to-end and user requests can communicate with the appropriate specialized AI Agent/API.

Milestone 4 – Complete QA, Deployment- $300

Scope

- Full end-to-end testing
- Registration/login testing
- Role-based flow testing
- Google/Apple authentication testing
- Image upload testing
- Database testing
- API testing
- Existing AI Assistant testing
- Individual AI Agent testing
- Agent-routing testing
- AI error/fallback testing
- Validation testing
- Edge-case testing
- UI/backend synchronization testing
- Final bug fixing
- Production environment verification
- Deployment support
- Source-code handover
- Technical handover

Deliverable

A stable, tested, and production-ready application with frontend, backend, database, APIs, authentication, storage, and supported AI Agents working together reliably.

The original QA milestone already covers authentication, role functionality, image uploads, AI functionality, APIs, databases, validation, edge cases, deployment, and handover; the updated version adds dedicated multi-agent testing.

7. Project Budget

Total Fixed Development Budget: $1,800

- Milestone 1: **$300**
- Milestone 2: **$600**
- Milestone 3: **$600**
- Milestone 4: **$300**

This keeps the same total and milestone structure as the existing proposal.